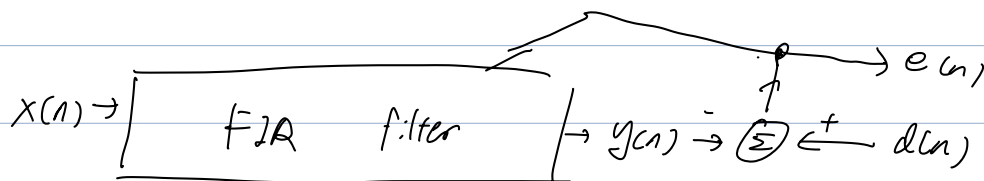


Adaptive FIR filter



$$e(n) = d(n) - y(n)$$

desired output

how can $e(n)$ change
FIR's coefficients?

gradient descent!

$$\Delta h_m = -\mu \cdot \frac{\partial (\frac{1}{2} e(n)^2)}{\partial h_m}$$

$$= -\frac{1}{2} \mu \cdot \frac{\partial (d(n) - \sum_{m=0}^{M-1} h_m x(n-m))^2}{\partial h_m}$$

$$= -\mu (d(n) - \sum h_m x(n-m)) \cdot (-x(n-m))$$

$$= \mu \cdot e(n) \cdot x(n-m)$$

$e(n) \rightarrow 0$ but not actually, to be zero.

it will contain any signal component
that are not correlated with $x(n)$.

$\therefore$ if we let $x(n)$ to be noise,

$\therefore e(n)$ is the cleaned up output signal.

eg. $d(n) = \text{signal} + \text{noise}$

$x(n) = \text{noise.}$



$e(n) = \text{signal.}$